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Smart library themes and elements: A systematic literature review

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Abstract

In recent years, libraries worldwide have been pushing to undergo drastic changes. These transformations impact library functions from traditional to electronic, digital, hybrid, virtual and smart libraries. However, there need to be more studies that systematically examine the elements that contribute to a smart library. Therefore, this article conducted a systematic literature review on the elements contributing to the smart library as a whole rather than focusing on one particular aspect. This study aims to explore the concept of smart libraries, including their essential elements, characteristics and theoretical foundations, to contribute to a deeper understanding of their role in transforming libraries. The review for this study was based on the publication standard and integrated multiple research designs using the review protocol RepOrting standards for Systematic Evidence Syntheses (ROSES). For this study, articles were selected from two major databases, Scopus and Web of Science, and a supporting database, Google Scholar. In these databases, 19 related studies were found on elements that comprise a smart library. The study's search strings excluded publication years to focus on identifying discussions about smart library elements. Results indicated publication dates spanning 2018–2022. Six major themes emerged from the thematic analysis of this review, including smart technology, smart services, smart people, smart governance, smart places and smart buildings. The study makes several significant contributions to practical purposes and the body of knowledge. The results explain how the elements can contribute to a smart library. The influence of ethical values in smart library elements and their contribution to library sustainability should be the subject of future research.

Keywords

Ethical value, smart library, smart library element, sustainability, systematic review

Introduction

'Library' is a familiar term in our society. The Oxford Dictionary defines a library as a building where collections of books, CDs, newspapers, etc., are kept for people to read, study, or borrow. However, as knowledge and technology evolve, so do the functions and tasks of the library. The American Library Association (2019) defines the library as a collection of resources in various formats organised by information professionals or other experts that provide convenient physical, digital, bibliographic or intellectual access and offer targeted programmes and services with the goal of educating, informing, or entertaining another user to promote individual learning and advance society as a whole.

As part of the information revolution, libraries have transformed in terms of management, services and functions from the traditional library to the electronic, digital, hybrid, virtual and smart library. The term 'smart library', inspired by the idea of the 'smart city', is now increasingly

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used to describe a new vision for libraries (Hussain and Ahmad, 2021). According to Zimmerman and Chang (2018), the term 'smart library' was coined by Aittola, Ryhanen and Ojala in 2003, and librarians have been working to implement smart libraries in different ways ever since. Smart libraries are a new way to provide interactive and innovative services and a dynamic way to expand the user base. Cao et al. (2018) stated that smart libraries represent a new way to provide innovative, interactive services and a dynamic method to expand the user base. A similar definition is given by Adetayo et al. (2021), who mentions that a smart library is the application of intelligent computing technology to improve the intelligence, integration and efficiency of a library's essential infrastructure processes and services. The general literature on smart libraries makes the rather simplistic assumption that any library that uses state-of cutting-edge information technology and provides smart services from smart people is a smart library.

To ensure that the library remains relevant and meets user's needs, it is crucial to understand what a smart library is, and what its characteristics are so that library management can use this as a guide for developing a smart library. This study focuses on reviewing the factors that make a library smart.

Towards a systematic review of smart library's elements

A systematic review or systematic literature review (SLR) is an assessment of a clearly stated question that uses systematic and explicit methods to discover, select and critically appraise relevant research and collect and analyse data from studies included in the review (Shaffril et al., 2018). It involves research synthesis, which is the process of evaluating and synthesising the findings of a search for global evidence relevant to a particular topic or set of questions. The results of this search are used to guide practice, policy and, in some cases, future research (Munn et al., 2018). Key benefits of SLR include transparency in data collection and synthesis, leading to greater objectivity and reproducibility (Kraus et al., 2020).

Shaffril et al. (2020) argue that a systematic literature review (SLR) is the best approach to this topic because it is an organised and transparent process that allows researchers to answer a defined question. The SLR process involves a rigorous search strategy that covers multiple databases, and the protocol or plan is established before the review process begins. This ensures that the search is conducted in a consistent and reproducible manner and that the review process is transparent and can be replicated by other researchers. In addition, the SLR process includes the classification, selection and critical appraisal of previous studies to answer the formulated question. This ensures that the review is comprehensive and that the studies selected are of high quality. Therefore, the systematic approach is preferable to other approaches, such as scoping, integrative and others, as it provides a rigorous and transparent process that ensures the quality and comprehensiveness of the review.

Hand-searching, hand-picking, or snowballing are commonly used techniques in literature reviews (LR) for article searches. These searches could be more comprehensive, and in most cases, authors of LR review articles or papers that they only discovered and were able to download. Apart from that, common LR does not do any quality assessment, and every article or paper found is reviewed. Kraus et al. (2020) claim that this leads to bias and subsequently affects the quality of the following review.

Although numerous studies on smart libraries exist, efforts to systematically review these studies are still insufficient. There are previous studies that have looked at smart libraries. For example, Cao et al. (2018) conducted a study to conceptualise the smart library concept. In contrast, Gul and Bano (2019) investigated emerging and novel technologies that are brought together to build smart libraries. Both studies use an in-depth review rather than a systematic literature review as a research method that can provide strategies for a more systematic review of the existing literature. The systematic review includes information about the review process used so that others can duplicate the study, verify the analysis, or examine the generality (e.g. keywords used, articles selected) (Shaffril et al., 2020). Even though some studies try to systematically investigate the topic of smart library items, they only pay attention to a selected feature. For example, Amaral et al. (2020) focus only on the Internet of Things as a feature of a smart library.

This study aims to conduct a comprehensive systematic review of the elements or features that contribute to the establishment of a smart library. It also examines the definition of a smart library and presents a theoretical framework. This article aims to fill the knowledge gap and define the elements of smart libraries.

This section explains why a systematic review is conducted, while the second section explains the methodology and review process ROSES (RepOrting Standards for Systematic Evidence Syntheses). The third section examines and synthesises the scientific literature to find, select and evaluate relevant studies on smart library elements. The final section identifies future research objectives.

Methodology

Source reference in SLR formation

This study was guided by the ROSES review protocol. ROSES (RepOrting standards for Systematic Evidence Syntheses) was used as a guide in this study. The ROSES initiative aims to raise high standards in the conduct of systematic reviews and maps by increasing transparency

and facilitating quality assurance of systematic reviews (Haddaway et al., 2018). Based on this review protocol, the authors began their SLR by developing related research questions for the review. Following this, the author establishes the systematic search technique. This is divided into three phases identification, screening (using inclusion and exclusion criteria) and eligibility. The authors then proceed to assess the quality of the selected papers and explain the strategy used to ensure that the articles reviewed are of high quality. Finally, the authors explain how the data were abstracted for the review and how they were analysed and validated.

Formulation of research question

This study aims to explore the concept of smart libraries, including their essential elements, characteristics and theoretical foundations, to contribute to a deeper understanding of their role in the transformation of libraries. Research Questions Development Tools (RQDT) were not used in this study. This SLR was guided by the following overarching research question to ensure that the primary objective was met: RQ1-What are the elements of a smart library? RQ2-What exactly is a smart library? and RQ3-What is the smart library's theoretical foundation?

Systematic searching strategies

Identification, screening and eligibility are the three main steps of systematic search strategies (see Figure 2).

Identification. Identification is a method of diversifying or expanding keywords for research. The identification method consists of searching for synonyms, related terms and variations for the primary keywords of the study. Synonyms, related terms and variations of keywords can be identified using an online thesaurus, keywords from previous studies, keywords from databases such as Scopus and Web of Science, and by obtaining opinions and statements from experts in the field. In this study, keywords found in a previous study on smart libraries were used. These keywords include element, feature, dimension and component of smart library. The goal is to give the selected database more options to find more relevant items for the study. The databases selected were Scopus and Web of Science, the most comprehensive databases of abstracts and citations from peer-reviewed literature (Shaffril et al., 2018). It was conducted in November 2022. The additional database chosen as an alternative database was Google Scholar. Some researchers may prefer to use Google Scholar because it is freely available and does not require institutional access or subscription fees besides can help researchers to identify additional relevant studies that may have been missed in their initial searches of Scopus and Web of Science.

The search string combinations use phrase search and truncation functions to develop complete keywords such as 'smart library', element, component, concept and dimension. In this study, the Boolean operator 'OR' was used to extend the search to the smart library features, and the Boolean operator 'AND' was used to include the smart library in the search. In this study, word binders or double quotes (' ') were used for the smart library phase to match terms in the phase search, and to allow searching based on verbs, derivatives, compounds, or variations, the truncation asterisk (*) was used to retrieve all possible articles about a library. The TITLE-ABS-KEY field is a searchable field in Scopus that includes the article title, abstract and author-provided keywords. Through searching within this field, researchers can increase the precision of their search results by focusing on articles that have relevant keywords in the title, abstract, or author-provided keywords. In this study, the use of the TITLE-ABS-KEY field code function allowed the researchers to narrow their search to articles that had the relevant keywords in the most important fields for indexing and searching. The combination of phrase search, truncation and the use of the TITLE-ABS-KEY field code function likely helped to produce a more precise and relevant set of search results (see Table 1). A total of 839 articles were retrieved by searching the three databases Scopus, Web of Science and Google Scholar.

Screening (inclusion and exclusion criteria). Screening is the process of selecting appropriate and SLR-relevant items based on established criteria, which is performed automatically by the database. Articles are screened before inclusion or exclusion based on the criteria selected by the authors, which is done automatically based on the sorting function of the database. During the screening process, inclusion criteria were used to limit and identify relevant papers to be selected for the following systematic review process. Document types were selected first as inclusion criteria; articles were then selected. Other forms of documents, such as conference papers, conference proceedings, reviews, book chapters, editorials and response papers, were excluded (see Table 2). Johnson and Hennessy (2019) emphasise the importance of selecting relevant document types such as original research articles, randomised controlled trials and meta-analyses to ensure the reliability and validity of the systematic review. Excluding other types of documents can reduce the risk of bias and improve the quality of the review. Therefore, the selection of document types is a crucial step in the systematic literature review process.

The second criteria for inclusion and exclusion purposes were language. In this study, only English-language documents were included, and all non-English-language documents were excluded to avoid misunderstandings and difficulties in translation in this study (see Table 2). Linares-Espinós et al. (2018) acknowledged that there may

Table 1. The searching strings.

Database	Search string
Scopus	TITLE-ABS-KEY ((‘smart librar*’) AND (element* OR characteristic* OR dimension* OR component*))
Web of Science	TS=((‘smart librar*’) AND (element* OR characteristic* OR dimension* OR component*))
Google Scholar	‘smart library’ element component characteristic dimension

be relevant studies published in other languages, but due to the language barrier, they decided to limit their review to English language documents. They also explained that this is a common practice in systematic reviews to ensure consistency and quality of the data collected. After the screening process, a total of 699 articles were removed from the 839 articles identified for this study (see Figure 2).

Eligibility. The third process of the systematic searching phase is eligibility, which can also be considered as a second screening process. This process will once again examine all the selected articles (from the screening process earlier) to ensure that all the selected articles meet the set criteria and further ensure that all the articles are related to the study. Eligibility is a manual process in which content is included or excluded based on the criteria selected by the authors. The articles found were thoroughly reviewed during this process, and any articles that did not meet the requirements were excluded. Duplicate documents were eliminated before the selection process began. A total of 25 articles were identified as duplicate articles that were excluded from both databases for the next phase, leaving 115 articles for the eligibility process that were manually screened for literature focused on the Smart Library elements and concept and the criteria from the earlier screening process (inclusion and exclusion criteria). The first step is to read the title of the article and use the title to assess whether or not it can be included in this SLR work. Second, if the title of the article needs clarification, read the abstract to determine if the article is appropriate. Finally, if the title of the article and the abstract remain unclear, read the entire article. Following this procedure, only 19 articles were selected for the next stage of the quality assessment process (see Figure 2). This process will once again examine all the selected articles (from the screening process earlier) to ensure that all the selected articles meet the set criteria and further ensure that all the articles are related to the study.

Article quality appraisal

Unlike common LR, SLR places great emphasis on evaluating the quality of articles. All articles selected after the systematic search must first be verified for quality before being peer-reviewed. There are two ways to evaluate the quality of an article: quantitatively or qualitatively (Petticrew and Roberts, 2008).

Table 2. The inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
Document type	Article	Conference paper, conference review, review, book chapter, editorial and retracted
Language	English	Non-English

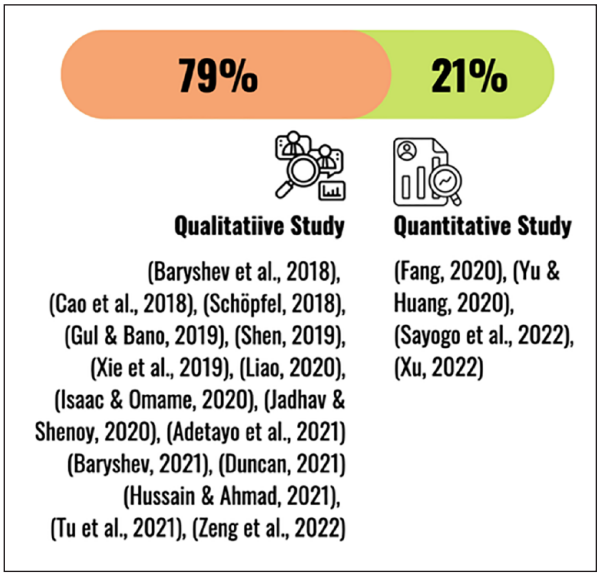


Figure 1. Article percentage based on methodology.

According to Hong et al. (2018), the Mixed Methods Evaluation Tool (MMAT) is a qualitative assessment tool that creates a critical appraisal tool for the appraisal phase of systematic reviews of mixed studies that include qualitative, quantitative and mixed methods research. It allows assessment of the methodological quality of studies in five categories: qualitative research, randomised controlled trials, nonrandomised studies, quantitative descriptive studies and mixed-methods studies. The selected articles indicated that a mixture of methods was used in these papers. Seventy-nine percent (15 articles) of the qualitative studies and 21% (4 articles) of the quantitative studies (see Figure 1). As a result, the authors chose to apply qualitative approaches to quality assessment using MMAT.

This study uses all the criteria categorised by the research method, that is, qualitative, with five criteria, and quantitative, with five evaluation criteria. In addition, two mandatory criteria of MMAT are also used to evaluate the selected articles. The criteria used are listed in Table 3 below.

Because critical appraisal is about making decisions, it is recommended that at least two reviewers participate independently in the appraisal process (Hong et al., 2018).

Table 3. Criteria of Mixed Methods Appraisal Tool (MMAT), version 2018 that applied in this study.

Category of study designs	Methodological quality criteria	Responses			
		Yes	No	Can't tell	Comments
Screening questions (for all types)	S1. Are there clear research questions?				
	S2. Do the collected data allow to address the research questions?				
	Further appraisal may not be feasible or appropriate when the answer is 'No' or 'Can't tell' to one or both screening questions.				
1. Qualitative	1.1. Is the qualitative approach appropriate to answer the research question?				
	1.2. Are the qualitative data collection methods adequate to address the research question?				
	1.3 Are the finding adequately derived from the data?				
	1.4. Is the interpretation of results sufficiently substantiated by data?				
	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?				
4. Quantitative descriptive	4.1. Is the sampling strategy relevant to address the research question?				
	4.2. Is the sample representative of the target population?				
	4.3. Are the measurements appropriate?				
	4.4. Is the risk of nonresponse bias low?				
	4.5. Is the statistical analysis appropriate to answer the research question?				

According to Shaffril et al. (2020), referring to Petticrew and Roberts (2008), experts should classify the remaining articles into three categories based on their quality: high, moderate and low. Only the articles in the high and moderate categories need to be reviewed. In evaluating the quality level, the experts focused on the methodology of the articles. Both reviewers must agree that the quality of the articles must be at least moderate to be included in the review. Before deciding whether to include or exclude certain articles from the review, they discussed any disagreements among themselves. After quality assessment by the reviewers, one article was rated as moderate, while 18 articles were rated as high. All 19 articles were thus eligible for review. Figure 2 illustrates the process from research question formulation to data extraction and analysis.

Data extraction and analysis

The process of extracting or adopting relevant data from previous studies for review must be based on the research question. A targeted search was conducted for research that answered the proposed questions. For the analysis process, the qualitative synthesis method was used in this SLR work. Several types of qualitative analysis can be used in SLR, and thematic analysis was chosen. Thematic analysis is simply an analysis that aims to uncover patterns in previous research findings. The pattern can be identified by two basic characteristics: Similarities between previous studies or correlations between previous studies. Thematic analysis is a widely used and easily accessible method for analysing qualitative data in psychology and beyond. It also provides a flexible approach to identifying and analysing themes and patterns in qualitative data that can be applied to a range of research questions and settings (Braun and Clarke, 2006). Thematic analysis was performed using the traditional method by first reading the abstracts and then reading the full articles (in depth) to find relevant themes and subthemes to the Smart Library elements. Table 4 shows the results of the thematic analysis. A detailed discussion of the themes created will be provided in the findings and discussion.

Findings

The authors have managed to select 19 articles. Based on the thematic analysis, six themes were identified: smart technology, smart services, smart people, smart governance, smart place and smart building. The terminology used to explain the above themes is also analysed: dimension, component, element and characteristic (see Table 4). Table 4 presents the countries that have made contributions to the publication, encompassing a total of 10 countries: China, Nigeria, Russia, India, Jamaica, Indonesia, France, the United States, Taiwan and Pakistan. In particular, China, Nigeria, Russia and India emerge as the dominant contributors within this group. Figure 4 provides a visual representation of the publication percentage breakdown per country. The search strings for this study were not included the year of publication, as the author generally wanted to know in which year the smart library was discussed.

From the review, six themes emerged for the element of a smart library. The six themes are smart technology (15 articles), smart services (11 articles), smart people

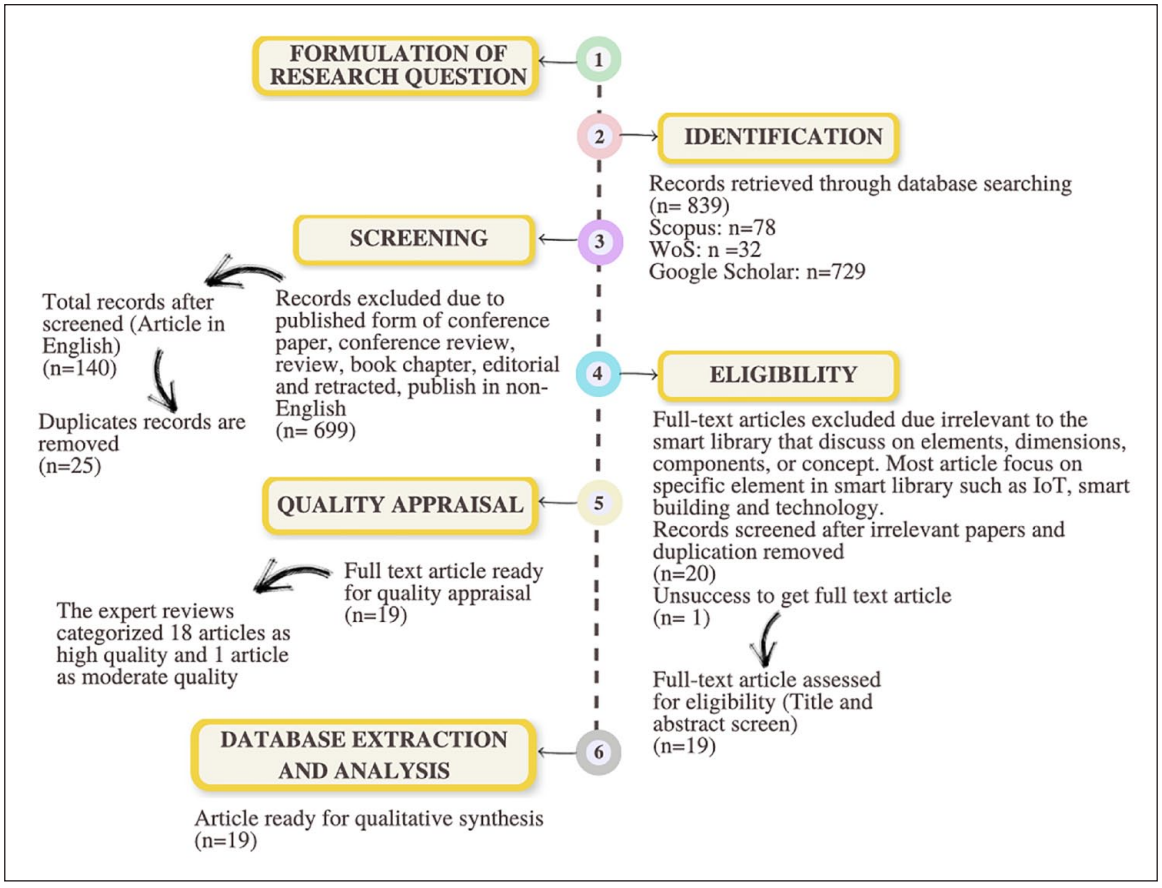


Figure 2. Systematic literature review flow diagram.

(6 articles), smart governance (4 articles), smart building (2 articles) and smart places (2 articles) (see Table 5).

A total of four studies used quantitative methods, while 15 other studies used qualitative methods in their studies. In terms of publication years, according to the results of this evaluation, publication starts in 2018 and ends in 2022, with three articles published in 2018, three in 2019, five in 2020, five in 2021 and three in 2022 (see Figure 3).

Discussion

In this study, several elements of the smart library were examined. Six themes emerged from the thematic analyses. This section discusses the developed themes in more detail. The number of papers based on thematic analyses is shown in Figure 4.

Smart technology

Smart libraries' main objective is to fulfil user information requests by utilising modern information technology. Smart technology in libraries can improve traditional and non-traditional library services, user experience and student learning opportunities (Baryshev et al., 2018; Isaac and Omame, 2020; Liao, 2020). Traditional libraries cannot be transformed into smart ones using a single technology. Instead, a diverse range of technologies must be integrated and interacted with. The smart library's basis and premise are advanced technology, including wearable, ubiquitous, mobile, intelligent processing, cloud computing and wearable technology. Fifteen of the study agreed that smart technology is one of the elements that needed to be included to form a smart library. Data mining, Internet of Things (IoT) and Artificial Intelligence (AI) are among the technologies that contribute to smart libraries (Cao et al., 2018; Gul and Bano, 2019; Zeng et al., 2022). Besides those three technologies, Hussain and Ahmad (2021) argue other smart technologies include Geographical Information Systems (GIS), Integrated Library Systems (ILS), Ambient intelligence (AI), Ambient Intelligence (sensor technology), Augmented Reality (AR), Blockchain technology, Radio Frequency Identification (RFID), Closed Circuit Television (CCTV) and Electronic Resources Management (ERM).

Fang (2020) argues that by integrating virtual reality technology into the smart library, readers gain access to a friendly, smart virtual environment that increases their willingness to use library services. Virtual reality technology is

Table 4. Thematic analysis (data extraction & data synthesis).

Studies (year)	Country	Term that used				Smart library elements					
		DI	CO	EL	CH	ST	SS	SP	SG	SC	SB
Adetayo et al. (2021)	Nigeria	√					√	√	√	√	
Baryshev (2021)	Russian				√		√				
Baryshev et al. (2018)	Russian					√	√				
Cao et al. (2018)	China	√	√			√	√	√			
Duncan (2021)	Jamaica	√				√	√				
Fang (2020)	China				√	√					
Gul and Bano (2019)	India		√			√	√	√			
Hussain and Ahmad (2021)	Pakistan			√		√	√	√			
Liao (2020)	China			√		√	√				
Isaac and Omame (2020)	Nigeria					√					
Jadhav and Shenoy (2020)	India			√		√	√	√	√		√
Sayogo et al. (2022)	Indonesia	√				√	√	√	√		
Schöpfel (2018)	France	√					√	√	√	√	
Shen (2019)	United States			√		√					√
Tu et al. (2021)	Taiwan			√		√					
Xie et al. (2019)	China			√		√					
Xu (2022)	China			√			√				
Yu and Huang (2020)	China					√					
Zeng et al. (2022)	China		√			√					

DI: dimension; ST = smart technology; SC: smart place; CO: component; SS: smart services; SB: smart buildings; EL: element; SP: smart people; CH: characteristic; SG: smart governance.

Table 5. Thematic analysis (Result based themes).

Author based themes	Total
Smart technology Baryshev et al. (2018), Cao et al. (2018), Duncan (2021), Fang (2020), Gul and Bano (2019), Hussain and Ahmad (2021), Liao (2020), Isaac and Omame (2020), Jadhav and Shenoy (2020), Sayogo et al. (2022), Shen (2019), Tu et al. (2021), Xie et al. (2019), Yu and Huang (2020), Zeng et al. (2022)	15
Smart services Adetayo et al. (2021), Baryshev (2021), Baryshev et al. (2018), Cao et al. (2018), Duncan (2021), Gul and Bano (2019), Liao (2020), Jadhav and Shenoy (2020), Sayogo et al. (2022), Schöpfel (2018), Xu (2022)	11
Smart people Adetayo et al. (2021), Cao et al. (2018), Gul and Bano (2019), Jadhav and Shenoy (2020), Sayogo et al. (2022), Schöpfel (2018)	6
Smart governance Adetayo et al. (2021), Jadhav and Shenoy (2020), Sayogo et al. (2022), Schöpfel (2018)	4
Smart building Jadhav and Shenoy (2020), Tu et al. (2021)	2
Smart place Adetayo et al. (2021), Schöpfel (2018)	2

a part of reality technology. It is characterised by strong interaction, integration of virtuality and reality and three-dimensional positioning. Meanwhile, Isaac and Omame (2020) conducted a significantly different study that focused on the use of social media and video conferencing in libraries. Without having to physically visit the four walls of the library, social media and video conferencing provide library users with direct access to the library and the librarian. The goal of smart libraries is to develop a mechanism to respond to and meet the needs and desires of their users. In the twenty-first century, libraries can leverage social media and videoconferencing to maximise their potential for effective service delivery by establishing social media profiles on Facebook, Twitter and WhatsApp and video conferencing facilities on Zoom, Skype and Google Hangouts for video conferencing with potential library patrons.

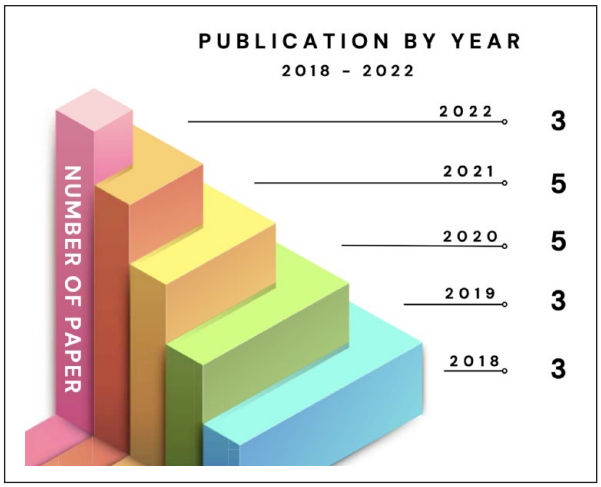


Figure 3. Number of paper by year.

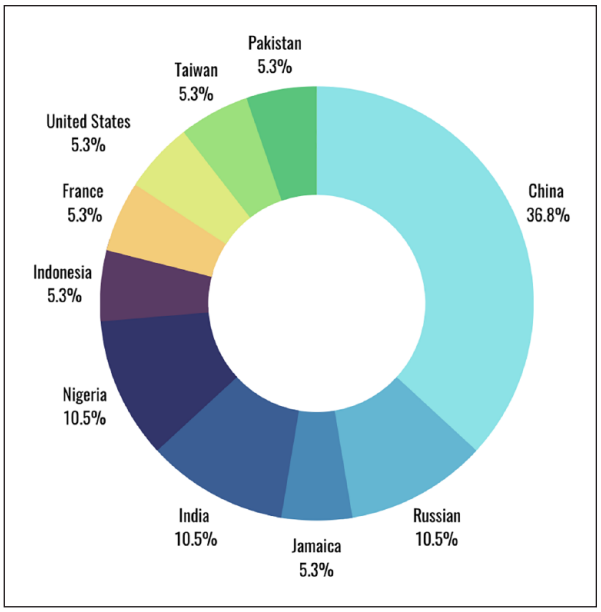


Figure 4. Percentages of publications by country.

Duncan (2021) also agrees that as smart technology advances, smart libraries will become smarter, increase their work capacity and satisfy users. The use of smart technologies in libraries would also bridge the gap between library services and the ever-changing and competing needs of users. RFID, cloud computing, artificial intelligence (AI), augmented reality (AR), Internet of Things (IoT), mobile internet, smart bookmarking apps, 5G and library electronic resource management are some of the emerging technologies that contribute to smart libraries. Shen (2019) suggested that libraries need to use AI to be relevant for two reasons: first, AI will enable better decision-making and efficient execution in data management, processing, extraction, summarisation and discovery. Second, AI can provide techniques for analysing data from distributed, autonomous sources and tools for identifying consistent knowledge states, resolving conflicts and making decisions. In addition, Yu and Huang (2020) also believe that the application of AI can help the library build services in a competitive environment and improve its resource offering to attract users.

Similarly, Xie et al. (2019) argue that by using RFID tags on books, the system can easily track material flow and visualise logistical activities. Ultimately, this improves the effectiveness of material flow, reduces the library's logistics costs, and contributes to the smartness of the library. Combined with RFID, the library should fully leverage the Internet of Things (IoT). Libraries are becoming smart, using networks of intelligent sensors and actuators connected to the Internet through IoT technology to provide a wide range of high-quality services to users and librarians. According to a study by Tu et al. (2021) on users' perceptions of smart libraries, they prefer learning environments with integrated technology in cultural and living exhibits, VR experiential spaces, interactive AI spaces, quiet and relaxing personal learning spaces and digital learning spaces. In addition, results show that consumers prefer to check out/return books and search for books on their own using computers, RFID technology, smartphones, tablets, self-service kiosks and smart bookshelves.

To measure the level of smartness of the library, Jadhav and Shenoy (2020) used three sub-elements of the technology used. First, the library employs cutting-edge technology to integrate and track all library collections and tagged items, including books, computers and devices. Second, the library uses technology to collect and preserve new digital objects. Third, the library enables its users, including those with print disabilities, to access the electronic content they want at any time and from anywhere in the world.

It can be concluded that the term smart technology refers to advanced technologies, including wearable technology, ubiquitous technology, mobile technology, intelligent processing, cloud computing and wearable technology, which contribute to the development of smart libraries by enhancing traditional and non-traditional library services, user experience and learning opportunities for students.

In addition to the benefits and changes that technology can bring to libraries, Sayogo et al. (2022) also identified challenges in the application of technology in the library. Smart libraries face two technological challenges: technological infrastructure and data. Two critical challenges libraries face in adopting a smart library are limited access to technology and the high cost of implementing rapidly changing technology. Critical issues in a smart library include librarians' limited ability to manage rapidly growing data, limited ability to use data innovatively and a lack of knowledge about the importance of data.

Overall, smart technology refers to the integration and adoption of advanced technologies, including IoT, AI, data

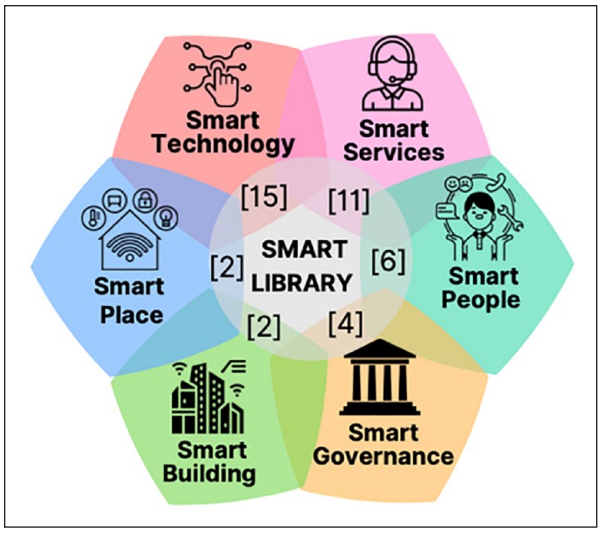


Figure 5. Number of Paper by Thematic Analysis.

mining, cloud computing and wearables, into traditional and non-traditional library services. The goal is to improve the user experience and learning opportunities by responding to information requests quickly and efficiently. Via using smart technologies, libraries can increase their work capacity, meet the ever-changing needs of users and reduce logistics costs while optimising resources and minimising waste (see Figure 5). However, implementing fast-changing technologies can be expensive and difficult, and librarians may need more training to manage and innovate with large amounts of data.

The convergence of blockchain technology and the Internet of Things (IoT) has the potential to reshape the landscape of library services (Khan et al., 2022a, 2022b). Thereby leveraging the security and transparency of blockchain and the practicality of IoT applications, libraries can usher in a new era of efficiency, security and user-centric experiences. This integration provides a comprehensive solution that addresses resource management, security issues and user engagement. As libraries adapt to the evolving technological landscape, the synergy between blockchain and IoT paves the way for a dynamic and responsive library ecosystem where seamless operations, improved resource access and enriching user interactions become the hallmarks of modern library services.

Smart libraries seamlessly integrate cutting-edge technologies to meet the changing needs of users and revolutionise service delivery, resulting in dynamic and user-centric environments. Emerging technologies such as the Internet of Things (IoT), electronic resource management (ERM), data mining, artificial intelligence (AI), ambient intelligence, blockchain and augmented reality are bridging the gap between library services and user expectations, increasing efficiency and satisfaction (Gul and Bano, 2019). This transformation is evident in 24/7 access to virtual resources, digitised collections with online catalogues, personalised recommendations, RFID-based collection management, interactive learning spaces, mobile apps, data-driven insights, immersive VR /AR experiences, digital preservation and seamless online learning collaboration. With leveraging these advances, smart libraries transcend time and space boundaries, provide seamless access to information and ultimately enhance traditional library services, user experience and student learning opportunities. Infrastructure integration, service building and people learning remain central to the development of these innovative smart libraries.

Smart services

A smart library is more than just an advanced technological application. It is also an improved service concept that provides excellent customer service and a great user experience. According to the information in about 11 relevant articles, the idea of smart library service includes a variety of service theories, such as service principle, service strategy and service mode, which form the smart service system and framework. Smart services that need to be applied in smart libraries include knowledge-based services, intelligent system-based services and high-level services that involve user participation (Cao et al., 2018). Meanwhile, Schöpfel (2018) and Adetayo et al. (2021) argue that smart library services are library-based ICTs, which are an adaptation of technological advances by using Big Data, IoT, AI, AR, wireless access, voice recognition and image recognition. Several university libraries have adopted various technologies to improve the efficiency of their daily services (Duncan, 2021; Gul and Bano, 2019). Applying these technologies to traditional library services will turn them into intelligent services.

Liao (2020) stated that smart services are strongly expressed and reflected in the improvement of the efficiency of library services through modern information technology. Smart libraries can also improve traditional and non-traditional library services, users' library experience and opportunities for student learning (Baryshev et al., 2018). By sourcing new information, monitoring global trends and using new technologies, librarians can develop creative services to meet users' needs (Yu and Huang, 2020).

Sayogo et al. (2022) conducted a study in Indonesia and found that the smart services offered by public libraries in Indonesia are divided into three categories. The first is smart programme offerings that use ICT (information and communication technologies), the second is greater programme reach and easier access to library programmes and services, and the third is user-centred programme offerings or value-added programmes such as entrepreneurship training.

To measure library intelligence, Jadhav and Shenoy (2020) included smart services as one of the measurement

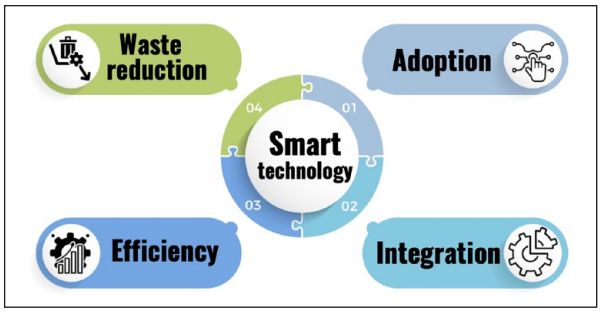


Figure 6. Smart technology characteristics.

elements, which includes four sub-elements. The elements are shown in Figure 6 below. The primary key to providing library services lies with library staff, especially librarians and is provided to users with different backgrounds and needs. The role of smart people in the smart library is discussed in the next section.

Based on the discussion held, the authors point out several commonalities and correlations regarding the dimensions of smart services in libraries. In particular, they all emphasise the importance of smart people as a critical component of a smart library, encompassing both library staff and users. They agree that the smart library should focus on improving the training and education of users in library use while at the same time improving the skills and knowledge of library staff in using smart technologies and serving smart users.

In addition, the authors emphasise that librarians need to develop smart skills such as information analysis, data mining, technology use, network security and knowledge acquisition to become well-trained professional smart librarians. They also emphasise the importance of lifelong learning and reflection for librarians to continuously improve their skills and knowledge.

Overall, smart services in a smart library involve the use of modern information technology to improve service efficiency, user experience and learning opportunities. These services can be categorised as knowledge-based, intelligent systems-based and user participation-based. Smart services are characterised by their user-centred, innovative and value-added approach. User-centred smart services aim to provide services that meet the needs and preferences of library users, with a focus on improving user training and education in library use. Innovative smart services involve the application of advanced technologies such as Big Data, IoT, AI, AR, wireless access, voice and image recognition to enhance traditional library services and develop creative services that meet user needs. Value-added smart services aim to provide users with additional benefits beyond traditional library services, such as entrepreneurship training (see Figure 7).

To ensure future success and improve service quality, libraries must incorporate cutting-edge technologies

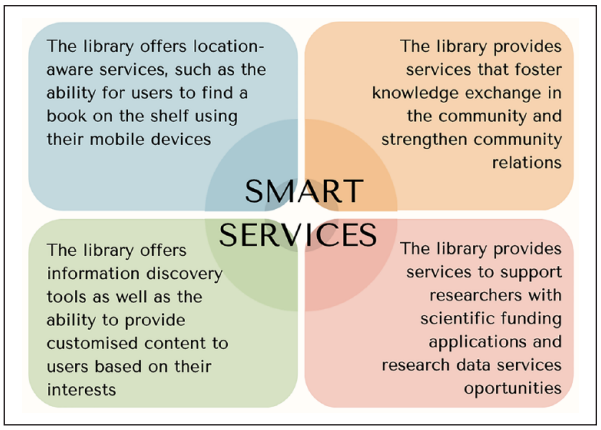


Figure 7. Element of smart services to measure library smartness according to Jadhav and Shenoy (2020).

strategically, prioritise user-centric approaches through continuous engagement and feedback, equip staff with smart skills, and cultivate a culture of innovation. In addition, they should provide user training, forge partnerships, offer value-added services and maintain adaptability to satisfy changing demands. Through implementing these principles, libraries can improve user experiences, maintain their relevance and flourish as intelligent, forward-thinking institutions.

Smart people

The smart people discussed in these studies are focused on both users and library staff. The smart library should improve the training of users in library use. The staff of a smart library should be knowledgeable as they handle smart technologies and serve smart users (Gul and Bano, 2019). Cao et al. (2018) stated in their study that an important human component of a smart library is the user and the librarian. The smart library must have a strategy to improve library education and user care. The results of the study indicate that user training requirements differ according to the three types of users: new users, users in urgent need of knowledge, and older users. In addition to users, librarians are also critical to managing the smart library. To become a truly smart library, the smart library needs well-trained professional smart librarians. The librarian education strategy includes developing librarians' smart skills and enhancing their awareness of lifelong learning by developing five skills for librarians: Information Analysis, Data Mining, Technology Use, Network Security and Knowledge Acquisition. In addition, librarians need to develop a lifelong learning mindset by reflecting on their own learning, combining self-improvement with smart library promotion, and learning proactively. Support from library leadership is important to encourage librarians to participate in developing their knowledge and skills.

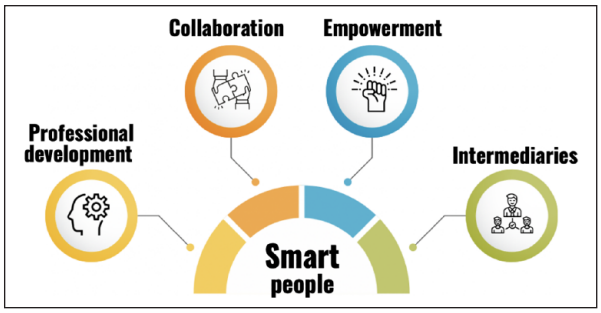


Figure 8. Smart people characteristics.

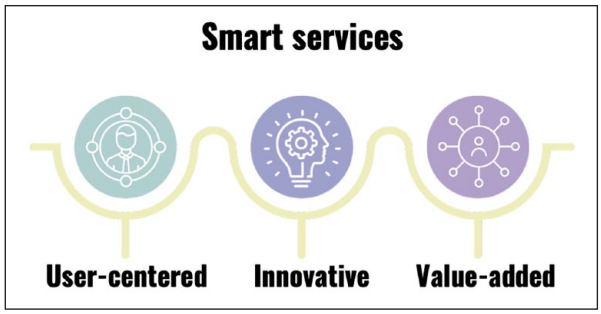


Figure 9. Smart services characteristics.

Adetayo et al. (2021) and Schöpfel (2018) also agree that the concept of smart people includes not only users but also library staff, their skills and professional development. The engagement of skilled librarians and users in creating public capacity speaks to the smart people component of a smart library, where users and librarians both serve as active producers of knowledge. The study states that the concept of smart people in a smart library consists of smart community and knowledge production. Sayogo et al. (2022) agreed in their study that smart people involve the joint production of innovative, collaborative knowledge between users and librarians. However, the development of smart people is still in its infancy in Indonesia, with efforts to promote smart people focusing on capacity building among both librarians and users.

The findings of Cao et al. (2018) and Schöpfel (2018) were used by Jadhav and Shenoy (2020) as a sub-element to measure the library smartness of smart people. First, the library sub-element often has a programme to improve user education on library use. Second, library users are encouraged not only to consume information but also to actively co-produce knowledge. The final sub-element is the library staff, who are able to manage and respond to intellectual property requests. In addition to smart people, smart governance is also seen as a disruptive factor in the smart library.

Smart people in a smart library refer to both users and library staff who have the knowledge, skills and lifelong learning awareness to handle and benefit from smart technologies. The smart library should provide training for users and develop well-trained professional smart librarians who can actively produce knowledge and manage intellectual property requests. Smart people are a critical component of a smart library, and efforts to promote smart people focus on building the capacity of both librarians and users (see Figure 8).

The authors of these studies have similar views regarding the smart people dimension of smart library services. They all emphasise the importance of both users and library staff and their skills and professional development. They agree that the concept of smart people involves the co-production of innovative, collaborative knowledge between users and librarians and that users should not only consume information but also actively co-produce knowledge. They also emphasise the need for a strategy to improve library education and user support and the importance of well-trained professional smart librarians. The authors emphasise the importance of librarians developing a lifelong learning mentality and the need for support from library management to encourage them to participate in developing their knowledge and skills.

In conclusion, smart people in the context of smart libraries refer to both users and library staff who have the necessary knowledge, skills and awareness to effectively use and benefit from smart technologies. Empowerment is an essential element of smart people, as users and staff should be trained and supported to improve their skills and knowledge about library services. Intermediaries also play a role in the smart people dimension, as librarians act as intermediaries between users and smart technologies, guiding and supporting users in their search for information. Collaboration is another important aspect of smart people, as both users and librarians work together to produce innovative knowledge. Professional development is also crucial for Smart People, as librarians need to continuously develop their skills and knowledge to effectively manage and respond to intellectual property requests and provide quality services to users (see Figure 9). Overall, smart people in smart libraries emphasise the importance of empowering both users and librarians through collaboration and professional development to create a knowledgeable and skilled community.

Smart governance

The next finding of this study is smart governance, which includes library governance, such as decision-making, sharing, public and collaboration. Adetayo et al. (2021) connect the smart library and smart governance through the nature of decision-making in library management. The study argues that the use of technology in library governance will contribute to a smart library. For example, the use of Big Data and text mining methods will help in decision-making and can be used for forecasting and better management of the library.

Meanwhile Jadhav and Shenoy (2020) and Schöpfel (2018) have presented different perspectives on smart governance. Key stakeholders need to have a clear understanding of the smart library they are promoting and its importance to the growth of their communities. User and library leadership involvement is essential to achieving the smart library vision.

Sayogo et al. (2022) stated that smart library management in Indonesia has two components. The first is user participation in decision-making based on user comments and contributions to improve library management. Second, collective intelligence provides creative programmes through collaboration between libraries and other institutions.

Jadhav and Shenoy (2020) used the findings from Schöpfel (2018) and divided them into three sub-elements to evaluate the smartness of the library through smart governance. First, the future of the library in terms of community sustainability is clearly defined by key stakeholders. Second, library governance relies on shared responsibility and collective intelligence among library staff, the library community and other institutions. Third, library management can identify sponsors and funding sources for new digitisation projects. Providing places and buildings that have certain characteristics will contribute to the smartness of libraries. However, there are not many studies that address these elements, including smart places and buildings.

Smart governance can be defined as a form of administration that uses technology and collaborative decision-making processes to improve the efficiency, effectiveness and sustainability of public services. In the context of library governance, smart governance involves the use of technology to facilitate decision-making, information sharing, public participation and collaboration among library stakeholders. The implementation of smart governance strategies in libraries can lead to the creation of smart libraries characterised by efficient and effective management practises, the use of data-driven decision-making processes and the provision of innovative services that meet the changing needs of library users.

The authors agree that smart governance is an essential component of a smart library. They also acknowledge that technology plays a crucial role in library governance, and that the use of Big Data and text mining methods can help in decision-making and improve library management. Furthermore, the authors emphasise the importance of user participation in decision-making and collective intelligence in library governance. In summary, the authors share similar views on the importance of smart governance in a smart library and agree that user participation, collective intelligence and technology are essential components for achieving the vision of a smart library.

Smart governance and smart people are closely related, since the success of smart governance depends on the skills and abilities of the people who administer it. This is a good reason to believe that smart governance can be combined with smart people. Smart governance requires leaders who are knowledgeable, innovative and have a deep understanding of their roles and responsibilities. Similarly, smart people are those who have the necessary skills and knowledge to drive innovation and progress in governance. They have a strong sense of civic responsibility and are able to work with others to achieve common goals.

In the context of smart libraries, smart governance means using technology and data-driven approaches to improve the efficiency and effectiveness of library services. Smart people in this context are those who are able to identify and implement these technologies and data-driven approaches to improve the user experience and better serve the community. They also have the skills to manage and analyse data to make informed decisions and continuously improve library services.

Smart place and smart building

As part of this study, two articles discuss a smart place and smart buildings. As argued, there are three dimensions of a smart place (Schöpfel, 2018). The first is an ecology that applies in parallel with the concept of a green campus. It includes aspects such as waste management, attractiveness of natural conditions, absence of pollution and sustainable management of resources. The second dimension is smart living, which refers to physical buildings that apply building management practices, such as safety, power and environmental management, as well as building improvements to attract interest and increase library use. A third dimension is a smart place that combines innovative green library features. It describes the transition of the conventional library structure and operation into a smart location that enhances the sustainability and smartness of the city. Adetayo et al. (2021) also agreed with the third dimension, which states that smart places combine services, users and the environment to improve the smartness of the library. The term ‘smart place’ combines new features of a ‘green library’ to describe the transformation of traditional library architecture and operation into a ‘smart place’ that contributes to the viability and intelligence of the city.

A smart place is one that integrates sustainable practices, building management strategies and innovative green library features to improve the intelligence and livability of the city. It encompasses three dimensions: Ecology, smart living and innovative green library functions.

Authors generally agree on the three dimensions of Smart Place, which include ecology, smart living and innovative green library features. Schöpfel (2018) introduced these dimensions and Adetayo et al. (2021) agreed on the third dimension, which describes the transformation of traditional library architecture and operations into a smart

Table 6. Definitions of smart library.

Author	Definition
Cao et al. (2018)	A holistic approach that combines intelligent technology (especially sensory technology) with ongoing training for librarians and the development of library users to deliver individualised and multifaceted library services that can adapt to the demands of different users.
Gul and Bano (2019)	Smart libraries are not only a new route for rendering interactive and innovative services but also adynamic mechanism for increasing the user base. Smart libraries have three important components smart technologies, smart people and smart services.
Shah and Bano (2020)	Smart library is a hardware and software complex with a wide range of opportunities for searching and providing necessary information to virtual users according to their inquiries and requirements.
Adetayo et al. (2021)	The use of smart computing technologies to make a city's basic infrastructure processes and services more intelligent, integrated and efficient

place. The authors also emphasise the importance of sustainability and smartness in library design and operation. However, there are not many studies that specifically address smart place as a dimension of a smart library and there is a need for further research in this area.

Meanwhiles, a total of 2 out of 19 articles discuss a smart building. In general, the debate about smart buildings and smart places is almost identical. Jadhav and Shenoy (2020) measure smart buildings by building management (energy efficiency, waste reduction, clean technological equipment) and space provision (users can explore, learn and use cutting-edge tools).

Using the same measurement method as Jadhav and Shenoy (2020), Tu et al. (2021) found that few students associate the ideas of 'smart buildings' with 'smart libraries', which may indicate that they are less familiar with smart buildings. Most students still think of smart libraries as lending, return and study/reading areas, which shows that they do not think of them as smart buildings but rather focus on technological tools, activities and objects and services.

Both articles emphasise the importance of building management in terms of energy efficiency, waste reduction and the use of clean technological equipment. Additionally, they both suggest that smart buildings should provide spaces that allow users to explore, learn and use cutting-edge tools. One of the articles also notes that there is a need to increase awareness among students about the concept of smart buildings in the context of smart libraries. Overall, the discussion on smart buildings in relation to smart libraries is still limited, but there is a clear consensus on the importance of energy efficiency and providing cutting-edge tools and spaces for users. Overall, a smart building can be defined as a building that utilises advanced technology to manage and optimise its energy efficiency, waste reduction and technological equipment while also providing an interactive and cutting-edge space for users to explore, learn and use.

From the above discussion, there are similarities between the concepts of 'Smart Place' and 'Smart Building' lies in their shared focus on the use of advanced technologies to improve environmental sustainability, efficiency and user experience in the context of libraries. Both concepts emphasise the integration of innovative technologies to create smart and user-centred environments.

In the case of 'Smart Place', the focus is on integrating sustainable practices, building management strategies and green library functions to improve the city's overall intelligence and quality of life. This concept encompasses dimensions such as ecology, smart living and innovative green library functions, all of which contribute to a holistic and environmentally friendly library space.

On the other hand, a smart building refers to a building that uses advanced technologies to manage energy efficiency, waste reduction and clean technological devices, while providing interactive and state-of-the-art spaces for users to explore, learn and use. The focus here is on smart management of the building's resources and facilities and creating an engaging and technologically advanced environment for users.

The similarity between these two concepts lies in the common goal of using technology to optimise physical spaces and resources while enhancing the user experience. Both 'Smart Places' and 'Smart Buildings' recognise the importance of integrating technology, sustainability and user-centred design to create modern, efficient and engaging library environments.

Definitions of smart library

As a consequence of the discussion regarding the smart library's elements, several authors have provided a definition of what a smart library is. Their definitions are provided in Table 6.

These definitions of a smart library all have in common the incorporation of intelligent technologies, flexible services and user-centred strategies. They place a strong emphasis on utilising technology, including hardware and software, to improve the delivery of information and services while taking into account the

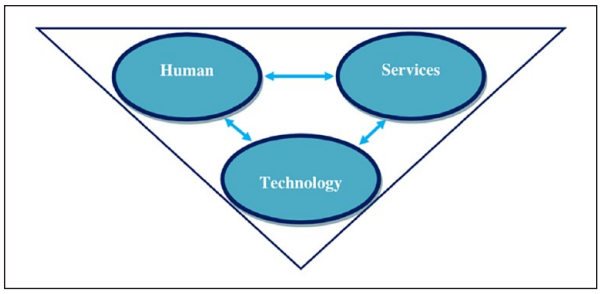


Figure 10. Framework 1 – Three basic elements of a smart library by Cao et al. (2018).

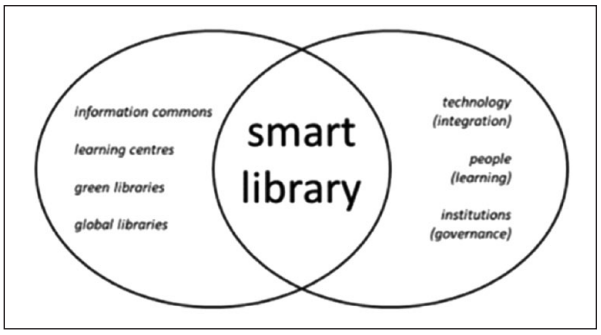


Figure 11. Framework 2 – Smart library at the interface between library concepts and the smart city by Schöpfel (2018).

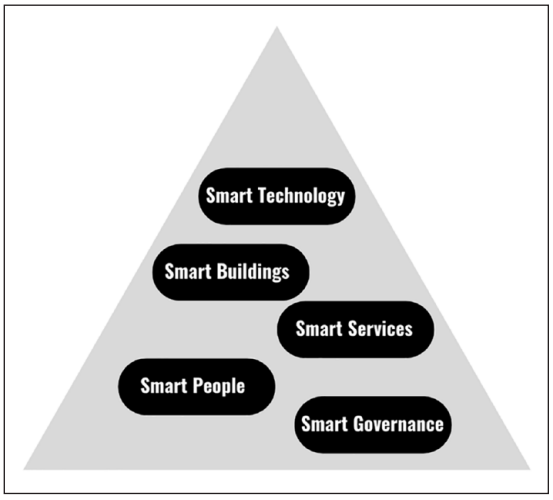


Figure 12. Framework 3 – Elements of smart library by Jadhav and Shenoy (2020).

preferences of specific users. Awareness of the larger context of efficiency and integration, either within the library itself or as part of a larger urban environment, as well as the importance of knowledgeable staff in leading and encouraging these interactions are also recognised. In general, it can be concluded that a smart library is an integrated and technologically cutting-edge organisation that combines intelligent technologies, user-centric services and skilled staff to deliver personalised and multifaceted experiences while also assisting in the integration and efficiency of urban services within the framework of a smart city.

Smart library theoretical foundation

Several writers have proposed frameworks for smart libraries, such as Cao et al. (2018), Schöpfel (2018) and Jadhav and Shenoy (2020). The framework is visually represented in Figures 10–12 provided below.

The analysis of the frameworks reveals common elements and themes in the conceptualisation of smart libraries. Technology, people and services are consistently identified as key elements, highlighting the importance of using advanced technologies, developing the skills and knowledge of library staff and providing innovative and user-centred services. The inclusion of governance and building elements in some frameworks further emphasises the need for effective management and physical infrastructure to support the implementation of smart libraries. Overall, these frameworks provide a comprehensive understanding of the key elements of the smart library concept and can serve as a foundation for further research and implementation in the field.

Recommendations

This study offers recommendations for future researchers in this field. According to the findings, a comprehensive smart library includes several key components: smart technology, smart services, smart people, smart governance, smart buildings and smart places. With this in mind, one possible avenue for further research is to group these six elements into three overarching categories: smart people, smart services and smart technology. Due to their inherent similarities, smart governance could be grouped under the smart people category, while smart place and smart buildings could be subsumed under the smart technology category.

An important observation is that the previous study did not adequately highlight the ethical management of all these elements. Future research should fill this gap by investigating the influence of ethical values on smart libraries to keep it sustain in the future. Ethics play a crucial role in ensuring the sustainability of libraries, especially in the context of their evolution into digital libraries. In the digital age, where technology and information are central to library operations, ethical considerations provide a framework that encourages responsible and transparent behaviour. Thus, establishing clear boundaries and principles for using and disseminating information, ethical management helps libraries navigate the complexities of smart technology, services and people. By adhering to ethical principles, libraries can foster an environment where information can be accessed, shared and preserved honestly. This not only improves the quality of service

delivery but also addresses issues on privacy, security and equal access. Integrating ethical principles into library operations ensures that smart libraries contribute positively to society, maintain their relevance, and are sustainable in an ever-changing technological environment.

Conclusion

This study's main purpose is to review the elements of smart libraries systematically. In this paper, several elements of the intelligent library have been studied. It can be concluded that technology is a major factor in making a library smart. However, other elements also contribute to a smart library, such as services, people, governance, building and place. Using cutting-edge technology is critical in assisting libraries in providing services to patrons. Using IoT Technology, Artificial Intelligence, Big Data, Mobile Apps and Augmented Reality in the service will allow management to make better, faster decisions. The usage of this technology is believed to be almost the same as the form of other libraries, such as 'digital library', 'electronic library' and 'virtual library'. However, the smart library focuses on more than just technology but also other elements such as smart people, smart services, smart governance, smart building and smart place. Those elements distinguish the smart library from other libraries.

The American Library Association (ALA) has established a code of ethics for librarians that emphasises intellectual freedom, privacy, access to information and professionalism. These ethical values are critical to the development of a smart library that uses technology and data to provide better services to users (American Library Association, 2021

In addition to ethical values, sustainability is also important in the development of Smart Libraries. Sustainability is about ensuring that the operation of the library is environmentally sound and that resources are used efficiently. This can include using energy-efficient technologies, reducing waste and promoting sustainable practices.

Combining ethical values and sustainability principles in the development of a smart library can result in a library that is not only technologically advanced but also socially responsible and environmentally conscious. This can improve the library's reputation, increase user satisfaction and contribute to the betterment of society as a whole.

Librarians play an important role in implementing plans and reforms in the library. Their knowledge and skills must match the current needs of library users. The concept of a smart library is not static; it will change according to the needs and changes of the world's industrial revolution. Changes in technology, civilisation and user intelligence will affect the concept of a smart library in the future. This systematic review study hopefully could facilitate researchers in providing a better understanding and emphasis of their information on the concept and elements of the smart library.

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References

Adetayo AJ, Adeniran PO, Gbotosho A, et al. (2021) Augmenting traditional library services: Role of smart library technologies and big data. *Library Philosophy and Practice*. Available at: https://digitalcommons.unl.edu/libphilprac (accessed 14 November 2022).

Amaral FV, Juliani JP and Bettio RWD (2020) Internet das coisas aplicada no ambiente das bibliotecas: Uma revisão sistemática da literatura internacional. *Perspectivas em Ciência da Informação* 25(4): 80–101.

American Library Association (2019) *LibGuides: Definition of a Library: General Definition*. The Librarian's Book of List. Available at: https://libguides.ala.org/library-definition/general

American Library Association (2021) Professional ethics. *Tools, Publications & Resources*. Available at: https://www.ala.org/tools/ethics

Baryshev RA (2021) From electronic to smart library systems: Concept, classification of services, scheme of work and model. *Journal of Siberian Federal University Humanities & Social Sciences* 14(3): 426–443.

Baryshev RA, Verkhovets SV and Babina OI (2018) The smart library project. *The Electronic Library* 36(3): 535–549.

Braun V and Clarke V (2006) Using thematic analysis in psychology. *Qualitative Research in Psychology* 3(2): 77–101.

Cao G, Liang M and Li X (2018) How to make the library smart? The conceptualization of the smart library. *The Electronic Library* 36(5): 811–825.

Duncan ASP (2021) Opportunities for academic smart libraries in the Caribbean. *Library Hi Tech News*. DOI: 10.1108/LHTN-06-2021-0035

Fang S (2020) Visualization of information retrieval in smart library based on virtual reality technology. *Complexity* 2020: 1–18.

Gul S and Bano S (2019) Smart libraries: An emerging and innovative technological habitat of 21st century. *The Electronic Library* 37(5): 764–783.

Haddaway NR, Macura B, Whaley P, et al. (2018) ROSES RepOrting standards for Systematic Evidence Syntheses: pro forma, flow-diagram and descriptive summary of the

plan and conduct of environmental systematic reviews and systematic maps. *Environmental Evidence* 7: 7.

Hong QN, Pluye P, Fàbregues S, et al. (2018) Mixed methods appraisal tool (MMAT) version 2018: User guide. Available at: http://mixedmethodsappraisaltoolpublic.pbworks.com/

Hussain A and Ahmad P (2021) Adoption of smart technologies in university libraries of Pakistan: A qualitative review. *Library Philosophy and Practice* 2021: 1–11. https://digitalcommons.unl.edu/libphilprac

Isaac AU and Omame IM (2020) Application of Social Media and Video Conferencing in Smart Library Services. *Library Philosophy and Practice*. Available at: https://digitalcommons.unl.edu/libphilprac

Jadhav D and Shenoy D (2020) Measuring the smartness of a library. *Library & Information Science Research* 42(3): 101036.

Johnson BT and Hennessy EA (2019) *Systematic reviews and meta-analyses in the health sciences: Best practice methods for research syntheses. Social Science & Medicine (1967)* 233: 237–251.

Khan AU, Zhang Z, Chohan SR, et al. (2022a) Factors fostering the success of IoT services in academic libraries: A study built to enhance the library performance. *Library Hi Tech* 40(6): 1976–1995.

Khan AU, Zhang Z, Taleby Ahvanooey M, et al. (2022b) Opinion mining towards blockchain technology adoption for accessing digital library resources. *Aslib Journal of Information Management* 74(1): 135–157.

Kraus S, Breier M and Dasí-Rodríguez S (2020) The art of crafting a systematic literature review in entrepreneurship research. *International Entrepreneurship and Management Journal* 16: 1023–1042.

Liao J (2020) The theoretical framework of library smart service literature review and content analysis. In: *Proceedings of the ACM/IEEE joint conference on digital libraries*, pp.471–472. DOI: 10.1145/3383583.3398569

Linares-Espinós E, Hernández V, Domínguez-Escrig JL, et al. (2018) Methodology of a systematic review. *Actas Urologicas Espanolas* 42(8): 499–506.

Munn Z, Peters MDJ, Stern C, et al. (2018) Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC Medical Research Methodology* 18(1): 143.

Petticrew M and Roberts H (2008) *Systematic Reviews in the Social Sciences: A Practical Guide*. Hoboken, NJ: John Wiley & Sons.

Sayogo DS, Yuli SBC and Wiyono W (2022) The determinants of Smart Public Library roles in promoting open government in Indonesia. *The International Information & Library Review* 54(2): 155–170.

Schöpfel J (2018) Smart libraries. *Infrastructures* 3(4): 43.

Shaffril HAM, Ahmad N, Samsuddin SF, et al. (2020) Systematic literature review on adaptation towards climate change impacts among indigenous people in the Asia Pacific regions. *Journal of Cleaner Production* 258: 120595.

Shaffril HAM, Krauss SE and Samsuddin SF (2018) *A systematic review on Asian's farmers' adaptation practices towards climate change. The Science of the Total Environment* 644: 683–695.

Shah A and Bano R (2020) Smart library: Need of 21 st Century. *Library Progress (International)* 40(1): 1–11. https://doi.org/10.5958/2320-317x.2020.00001.x

Shen Y (2019) Emerging scenarios of data infrastructure and novel concepts of digital libraries in intelligent infrastructure for human-centred communities: A qualitative research. *Journal of Information Science* 45(5): 691–704.

Tu Y-F, Hwang G-J, Chen S-Y, et al. (2021) Differences between LIS and non-LIS undergraduates' conceptions of smart libraries: A drawing analysis approach. *The Electronic Library* 39(6): 801–823.

Xie Y, Liu J, Zhu S, et al. (2019) An IoT-based risk warning system for smart libraries. *Library Hi Tech* 37(4): 918–932.

Xu K (2022) *Intelligent Library Service and management based on IoT assistance and text recommendation*. *Journal of Sensors* 2022: 1–10.

Yu K and Huang G (2020) Exploring consumers' intent to use smart libraries with technology acceptance model. *The Electronic Library* 38(3): 447–461.

Zeng Z, Sun S, Li T, et al. (2022) Mobile visual search model for Dunhuang murals in the smart library. *Library Hi Tech* 40: 1796–1818.

Zimmerman T and Chang HC (2018) Getting smarter: Definition, scope, and implications of smart libraries. In: *Proceedings of the ACM/IEEE joint conference on digital libraries*, pp.403–404. DOI: 10.1145/3197026.3203906

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